

Impact of slow versus rapid digesting carbohydrates on substrate oxidation in pre-pubertal children: A randomized crossover trial

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ABSTRACT

Background & Aims: Consumption of rapid digesting sugars by children are under increased scrutiny because of their contribution to unhealthy weight gain. Previous studies in adults and children have suggested that altering the blend of carbohydrates (CHOs) consumed may cause shifts in substrate utilization. The purpose of this study was to examine the effects of consuming a slow digesting carbohydrate (SDC) and rapid digesting carbohydrate (RDC) on CHO and fat oxidation, glucose, and insulin responses at rest, during exercise, and post-exercise rest in pre-pubescent children. **Methods:** A randomized, double-blind, crossover design was used. Nineteen pre-pubescent children (n=10 boys, n=9 girls, mean±standard error, age=9.84±0.37-yr) participated. Visits to the laboratory began with a 30-min measurement of resting metabolism followed by consumption of either an RDC or SDC drink. Postprandial resting metabolism was recorded for 60-min, immediately followed by 60-min of submaximal cycling exercise while metabolism was recorded, which was immediately followed by another 60-min recording of post-exercise metabolism. Total CHO and fat oxidation, endogenous and exogenous CHO oxidation, blood glucose, and insulin were assessed. **Results:** Total CHO oxidation rate ($\text{g}\cdot\text{min}^{-1}$) was greater after the RDC drink at 60 min ($p=0.032$). Endogenous CHO oxidation rate ($\text{g}\cdot\text{min}^{-1}$) was greater after the SDC drink at 15 min ($p\leq 0.010$). Cumulative endogenous CHO oxidation (g) was greater after the SDC drink at 45 min ($p=0.009$). Endogenous CHO oxidation accounted for a greater proportion of substrate oxidation after the first 60-min rest period ($p=0.028$), while exogenous CHO oxidation accounted for a greater proportion of substrate oxidation for the RDC at all time points ($p\leq 0.019$). **Conclusions:** The present study provides novel data suggesting that an SDC promotes greater endogenous substrate utilization in pre-pubertal children, which may

have beneficial health impacts on energy intake and carbohydrate regulation/metabolism during growth and development.

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Keywords: children; carbohydrate oxidation; fat oxidation; endogenous; exogenous

Abbreviations: CHO (carbohydrates), $\dot{V}O_2$ max (estimated maximal oxygen uptake), RDC (rapid digesting carbohydrate), SDC (slow digesting carbohydrate), WHO (World Health Organization)

1. INTRODUCTION

Consumption of rapid digesting sugars, particularly in the absence of physical activity, by children are under increased scrutiny because of their contribution to unhealthy weight gain. Accordingly, there are aggressive efforts to reduce sugar consumption while increasing physical activity levels in children on a global scale. In fact, the World Health Organization (WHO) [1] published guidelines recommending that rapid digesting sugars should contribute to less than 10% of a child's total energy intake. The scientific rationale for restricting intake of rapid digesting sugars appears to be strongly related to their extreme palatability, which may promote overeating and positive energy balance and lead to increased fat accumulation, insulin resistance and obesity, weight gain and increased risk of future obesity-related comorbidities [1–5]. In children, randomized-controlled trials [6,7] demonstrated that fat weight gain and/or increases in body mass index over 6 months to 2 years were slowed when sugar-sweetened beverages were replaced with sugar-free, lower calorie beverages. In addition, a recent study in adults demonstrated that consumption of slow digesting sugars such as isomaltulose, an isomer of sucrose linking fructose and glucose by an alpha-1,6-glycosidic bond, with lower glycemic index (32 for isomaltulose versus 64 for sucrose) may help manage body weight by inducing greater body weight loss as fat mass compared to consumption of sucrose [8]. The authors speculated that the higher overall fat oxidation at the expense of carbohydrate (CHO) oxidation may account for the beneficial effects of isomaltulose [8]. Furthermore, two studies [9,10] in pre-adolescent children showed that the amount of food consumed at lunch was greater when high glycemic CHOs were consumed at breakfast. The collective results of these previous studies suggest that consumption of different types of CHO (i.e., rapid versus slow digesting CHO) may impact not only glycemic response, but subsequent energy intake, as well as fat and carbohydrate oxidation.

In animals, previous studies have suggested that consumption of slow digesting CHO (SDCs) may promote greater metabolic flexibility during the later stages of growth and development [11–14]. In fact, a diet consisting of rapid digesting CHO (RDCs) demonstrated increased lipogenesis and subsequent fat mass percentage, total cholesterol, and triglycerides during the early stages of growth and development [11,13,14], while a similar diet consisting of SDCs demonstrated a healthier progression of these markers, and in fact demonstrated greater development of lean mass [12,13]. Furthermore, in animals, consumption of SDCs may improve muscle functionality [12,13], reduce predisposition for non-alcoholic fatty liver disease [11,13], and combat obesity from growth and development to the later stages in life [13,14]. Although less is known regarding post-prandial metabolism after consumption of SDCs and RDCs in human subjects, previous studies in adults and children have suggested that altering the chemical composition of CHO consumed, such as the glycemic index, during periods of rest and exercise may cause shifts in substrate utilization during submaximal exercise [15–22]. Since both composition of CHO consumed and exercise may cause shifts in substrate utilization it is crucial to understand the influence of SDCs and RDCs on resting and exercise metabolism to provide better recommendations on how the composition of CHO consumed in combination with exercise may aid in combatting unhealthy weight gain in children. However, we are unaware of any previous studies to specifically compare the influence of SDCs and RDCs on substrate utilization during rest, exercise, and post-exercise rest. Thus, the purpose of this study was to examine the effects of consuming an SDC and RDC on postprandial CHO and fat oxidation, glucose, and insulin responses at rest, during exercise, and post-exercise rest in pre-pubescent children.

2. MATERIALS AND METHODS

2.1. Participants

Based on the results of previous studies [15–22], which used sample sizes ranging from $n = 8$ to $n = 12$, and a priori calculations of sample size based on the equation from Vincent and Weir [23], it was determined the minimum number of participants needed to detect changes in carbohydrate (total, endogenous, and exogenous) and fat oxidation was $n = 15$. Therefore, the proposed sample size was $n = 20$, with an anticipated attrition of 25% to complete the study with $n = 15$ eligible participants. Forty-five participants were pre-screened for eligibility. Of those pre-screened, $n = 21$ were enrolled in the study. One participant withdrew prior to the Baseline Visit and one participant withdrew after the Baseline Visit, but before Test Visit 1. Therefore, nineteen pre-pubescent children ($n = 10$ boys, $n = 9$ girls, mean $\pm$ standard error, age = 9.84 ± 0.37 years, maturity offset = -2.91 ± 0.18 years, height = 142.54 ± 2.49 cm, body mass = 35.76 ± 2.14 kg) were included in this study. Figure 1 provides a flow chart for participant recruitment.

Participants were included in this study if they met the following inclusion criteria: a) male and 9-13-years old, or female and 8-10-years old; b) Tanner stage 1 based on participant self-report; c) born full-term, defined as 37-40 weeks; d) born with normal body mass, defined as 2.5-4.3 kg; e) answered “no” to questions 1-7 or all of the follow-up questions of the Physical Activity Readiness Questionnaire (PAR-Q+ 2015); f) body mass index $> 5^{\text{th}}$ percentile and $< 95^{\text{th}}$ percentile; and g) provided signed youth assent and signed parental consent as approved by the University of Nebraska Medical Center Institutional Review Board (IRB# 167-17-FB, title: *AL25 Proof of Concept Study: Carbohydrates and Children*). Participants were excluded from this study if they met the following exclusion criteria: a) parent stated the participant had a history of metabolic, endocrine, or hyperactivity disorders; b) parent stated the participant was known to be allergic or intolerant to any foods including ingredients found in the study products; or c) parents

stated the participant was participating in a concomitant trial of a non-registered drug (or is within the 30-day follow-up period for such a trial) or that otherwise conflicted with this study.

[FIGURE 1 ABOUT HERE]

2.2. Study Design

A randomized, double-blind, crossover design was used. Figure 1 provides a flow diagram for assessment, randomization, allocation, and analysis of study participants. Participants completed four study visits: one Pre-screening Visit, one Baseline Visit, and two Test Visits. The order of visits is visually depicted in Figure 2. The Pre-screening Visit was performed by the investigators via telephone or during an in-home visit to assess participant eligibility. Seven to 14 days after the Pre-screening Visit, eligible participants came to the laboratory in the morning after an 8-hour fast for the Baseline Visit. During the Baseline Visit, anthropometrics, body composition assessments, and a submaximal cycle ergometer test were performed. At the end of the Baseline Visit, delivery of the pre-packaged foods to be consumed for the three days prior to Test Visit 1 was scheduled. For both Test Visits, participants reported to the laboratory in the morning after an 8-hour fast, 7-14 days after the preceding visit. Figure 2 visually depicts the Test Visits, which began with an initial 30-min measurement of resting metabolism followed by consumption of either an RDC or SDC drink. Participants who were randomly assigned to receive the RDC for Test Visit 1 consumed the RDC after the initial 30-min period of resting metabolism (Figure 1 and 2). When they returned for Test Visit 2, those same participants received the SDC (Figure 1 and 2). Likewise, participants who received the SDC for Test Visit 1 then received the RDC upon Test Visit 2 (Figure 1 and 2). All participants were required to consume the drink (regardless of RDC or SDC) within a 10-min period. After consuming the drink, resting metabolism was recorded for 60-min, immediately followed by 60-

min of submaximal exercise metabolism, which was immediately followed by another 60-min recording of post-exercise metabolism. Throughout each Test Visit, open-circuit spirometry and indirect calorimetry were used to monitor oxygen consumption and substrate utilization. Breath tests were also performed every 15-min to track naturally labelled ^{12}C and ^{13}C in expired CO_2 . Blood samples were obtained for glucose and insulin. At the end of Test Visit 1, delivery of the pre-packaged foods to be consumed for the three days prior to Test Visit 2 was scheduled.

[FIGURE 2 ABOUT HERE]

2.3. Anthropometrics and Body Composition

Height, seated height, and body mass were measured using a calibrated digital scale and stadiometer (Seca 769, Hamburg, Germany). Maturity offset from peak height velocity was calculated based on the equations from Mirwald et al. [24]. Fat mass (kg) and fat-free mass (kg) were assessed using whole-body dual-energy x-ray absorptiometry (Lunar iDXA, GE Healthcare, Madison, WI). The dual-energy x-ray absorptiometry device was calibrated daily using a quality assurance phantom that consisted of varying bone mineral density and percent fat standards (Lunar iDXA User Manual, GE Healthcare, Madison, WI). Before each scan, the participant's height, body mass, sex, birthdate, and race were entered into the dual-energy x-ray absorptiometry software (EnCore, GE Healthcare, Version 14). Participants' bodies were free from any external metal objects, and no internal metal devices were noted. Participants were instructed to lie supine on the padded scanner table with their hands pronated and near their body, but without touching the hip. A Velcro strap was wrapped around the toes and ankles to keep the participant's legs adducted. The appropriate thickness (thin, standard, or thick) based on chest depth was selected prior to completing each total-body scan (Lunar iDXA User Manual,

GE Healthcare, Madison, WI). Descriptive outcome variables included fat-free mass, fat mass, and percent body fat.

2.4. Submaximal Estimation of $\dot{V}O_2$ max and Workload

All exercise tests were completed on an electronically-braked cycle ergometer (Corival cpet, part #960900, Lode B.V., Groningen, the Netherlands) with modified adjustable crank arms (part #928804, Lode B.V., Groningen, the Netherlands). Prior to beginning each test, the seat height of the cycle ergometer was adjusted so that the seat was at the height of each participant's anterior superior iliac spine. The seat height and crank arm settings were initially recorded at the Baseline Visit so that the same settings for each participant were replicated for Test Visits 1 and 2. During the Baseline Visit, participants completed a submaximal cycle ergometry test to estimate maximal oxygen uptake ($e\dot{V}O_2$ max). Throughout the test, participants were instructed to maintain a cadence of 50 revolutions per min, which was visually displayed on a screen. Each test began with the resistance of the cycle ergometer set at 25 W for 3-min. After the initial 3-min stage, the resistance on the cycle ergometer was raised to 75 W for 3-min. All subsequent 3-min stages increased the resistance on the cycle ergometer by 25 W. Throughout the test, heart rate was recorded with a wireless chest strap (Polar H10, Polar USA, Bethpage, NY) interfaced with the metabolic cart. Once the participant's heart rate reached 75% of the age-predicted maximum (defined as $220 - \text{age}$) [25], the test was terminated.

After the initial 3-min of the submaximal cycle ergometer test was eliminated to account for the cardiac adjustment period, linear regression was used to predict $e\dot{V}O_2$ max ($\text{ml}\cdot\text{kg}\cdot\text{min}^{-1}$) and the workload at $e\dot{V}O_2$ max (W). First, each participant's recorded $\dot{V}O_2$ (y-axis) was plotted over recorded heart rate (x-axis). A linear regression equation was fit to the data. Estimated $\dot{V}O_2$ max was predicted from each regression equation by solving for the age-predicted maximum

HR. Second, each participant's workload (W, y-axis) was plotted over recorded heart rate (x-axis). A linear regression equation was fit to the data. Estimated maximal workload (W) was predicted from each regression equation by solving for the age-predicted maximum heart rate.

During each Test Visit, participants completed 60-min of submaximal exercise on the same cycle ergometer used at the Baseline Visit (Figure 2). The resistance of the cycle ergometer was set at 50% of the workload corresponding with $\dot{V}O_2$ max. Throughout the test, participants were instructed to maintain a cadence of 50 revolutions per minute.

2.5. Oxygen Consumption and Indirect Calorimetry

During the submaximal cycle ergometry test at the Baseline Visit and during the resting and exercise periods at both Test Visits (Figure 2), expired air was analyzed with open-circuit spirometry and indirect calorimetry (Parvo Medics TrueOne® 2400 Metabolic Measurement System, Sandy, Utah). Before each use, oxygen and carbon dioxide gas analyzers were calibrated with a gas mixture of 4% CO₂, 16% O₂, and 80% N, and the spirometer was calibrated with a 3 L syringe (Series 5530, Hans Rudolph, Inc., Shawnee, KS) according to manufacturer instructions. Face masks (7450 V2, Hans Rudolph, Inc., Shawnee, KS, size extra small), headgear (7450 V2, Hans Rudolph, Inc., Shawnee, KS), and attached two-way non-rebreathing valves (2700, Hans Rudolph, Inc., Shawnee, KS) were secured around the nose and mouth of each participant to form an air tight seal on the face. Breath-by-breath measurements of $\dot{V}O_2$ and $\dot{V}CO_2$ were averaged over 15-min time epochs (Figure 2). Fat and CHO oxidation were calculated using standard stoichiometric equations [26]:

$$\text{Carbohydrate Oxidation (g}\cdot\text{min}^{-1}\text{)} = 4.55 (\dot{V}CO_2) - 3.21 (\dot{V}O_2) - 2.87 (n)$$

$$\text{Fat Oxidation (g}\cdot\text{min}^{-1}\text{)} = 1.67 (\dot{V}O_2) - 1.67 (\dot{V}CO_2) - 1.92 (n)$$

Where n represents urinary nitrogen excretion. In the above calculations, the carbohydrate and fat oxidations were based on a $\dot{V}O_2$ of $2.500 \text{ L} \cdot \text{min}^{-1}$ and a $\dot{V}CO_2$ of $2.250 \text{ L} \cdot \text{min}^{-1}$ and negligible protein oxidation ($n = 0$) [27]. Outcome variables from the indirect calorimetry included $\dot{V}O_2$, $\dot{V}CO_2$, fat oxidation ($\text{g} \cdot \text{min}^{-1}$), and CHO oxidation ($\text{g} \cdot \text{min}^{-1}$).

2.6. Breath Tests

Breath samples were collected by asking participants to breath into a plastic straw (VP116/C, Breath Sampling Straws, Labco Limited, Lampeter, UK) exiting into a 12 mL exetainer vial (439W, Exetainer® 12 mL coated vial-No Vacuum 439W, Labco Limited, Lampeter, UK). The vials were sealed and stored until analysis. A baseline breath sample of expired gas for analysis of ^{13}C abundance was collected before consumption of the CHO drink. All subsequent breath samples were collected every 15 minutes for the remainder of the Test Visits (Figure 2). All breath samples were analyzed via isotope ratio mass spectrometry (VG Optima duel inlet stable isotope ratio mass spectrometer, Isoprime Inc., Manchester, UK) for ^{12}C and ^{13}C abundances in expired CO_2 [28]. Calculation of the ^{13}C abundance was performed with the following equation [29]:

$$\Delta^{13}\text{C} = \left(\frac{{}^{13}\text{C}: {}^{12}\text{C}_{\text{sample}} - {}^{13}\text{C}: {}^{12}\text{C}_{\text{standard}}}{{}^{13}\text{C}: {}^{12}\text{C}_{\text{standard}}} \right) \times 1000 \text{ ‰}$$

Where $\Delta^{13}\text{C}$, in parts per million (‰), reflects the abundance of ^{13}C in a sample relative to the abundance of that same isotope in the reporting standard from Vienna Pee Dee Belemnite Formation limestone with a $^{13}\text{C}: {}^{12}\text{C}$ of 0.011237 [28]. Exogenous CHO oxidation ($\text{g} \cdot \text{min}^{-1}$), after consumption of the CHO drink, was calculated with the following equation [30]:

$$\text{Exogenous CHO Oxidation} = \dot{V}CO_2 \left(\frac{\Delta^{13}\text{C}_{\text{timepoint}} - \Delta^{13}\text{C}_{\text{baseline}}}{\Delta^{13}\text{C}_{\text{ingested}} - \Delta^{13}\text{C}_{\text{baseline}}} \right) \times \left(\frac{1}{k} \right)$$

Where $\Delta^{13}\text{C}_{\text{timepoint}}$ was the abundance of ^{13}C in the breath sample at each individual time point, $\Delta^{13}\text{C}_{\text{baseline}}$ was the abundance of ^{13}C in the baseline breath sample immediately prior to ingestion of the CHO drink, $\Delta^{13}\text{C}_{\text{ingested}}$ was the abundance of ^{13}C in the CHO drink, and k ($0.7467 \text{ L}\cdot\text{g}^{-1}$) is the volume of CO_2 produced when 1 g of glucose is completely oxidized. Endogenous CHO oxidation ($\text{g}\cdot\text{min}^{-1}$) was then calculated as the difference between total CHO oxidation and exogenous CHO oxidation for each time point.

2.7. Blood Samples

Capillary blood samples (200 μL) were collected in microvettes (Microvette® 200 μL , K3 EDTA, violet US code; 10.8mm x 46.6 mm, Sarstedt, Inc., Newton, NC) after lancet finger punctures taken directly prior to consumption of the CHO drink, then every 30-min during the initial 60-min rest, and every 60-min for the final 120-min of each Test Visit (Figure 2). Glucose ($\text{mg}\cdot\text{dL}^{-1}$) was measured with a handheld blood glucose monitoring system (Freestyle Freedom Lite®, Abbott Diabetes Care Inc., Alameda, CA) and recorded immediately. After each Test Visit, blood samples were centrifuged, and serum was pipetted and stored at -80°C until analysis. Serum samples were analyzed in duplicate by enzyme-linked immunosorbent assays (ELISAs) to determine concentrations of insulin ($\text{mU}\cdot\text{L}^{-1}$) (Insulin ELISA kit, Crystal Chem High Performance Assays) per the manufacturer instructions.

2.8. Carbohydrate Drinks

The RDC drink provided 30 g of CHO from a 50:50 ratio of maltodextrin:sucrose. Ingredients for the SDC drink included 7.8 g isomaltose (Beneo, Germany), 6.9 g maltodextrin, 6.6 g sucromalt glucerna (Cargill, Minneapolis, MN), 3.6 g vita fiber isomaltooligosaccharide (Vita Fiber, Bioneutra, Canada), 3 g resistant maltodextrin (Fibersol-2, ADM, IL), and 2.1 g inulin-oligofructose (Beneo, Germany). Ingredients were shipped to the laboratory in bulk from

the sponsor (Abbott Nutrition, Columbus, OH) and drinks were mixed at the study site in 237 mL distilled water and flavored with 4 g of sugar-free lemonade drink mix (Crystal Lite, Kraft Food, Glenview, IL) to mask differences in taste. The ingredients for each drink were stored in an access-controlled environment at ambient temperature within 1-hour before consumption and were weighed and mixed under the supervision of a registered dietician. The isotopic abundances of the drinks were assessed with mass spectrometry, as described in detail previously by Shoemaker et al. [31]. To maintain the double-blind study design, the registered dietician responsible for mixing the drinks was responsible for randomization of the participants, using a random sequencing generator, and was not directly involved with any participant interaction.

2.9. Pre-packaged Foods

Dietary intake for the three days prior to each Test Visit was standardized across participants to reduce the enrichment of expired CO₂ from naturally derived ¹³C. For the standardized diet, investigators chose thirty-six pre-packaged foods from a local grocery store that did not include ingredients derived from C₄ plants, such as corn and sugar cane [15]. Meal plans including pre-packaged foods were designed by a registered dietician and individualized to each participant's daily energy intake needs based on a three-day food intake record completed prior to the Baseline Visit. Participants were provided the pre-packaged foods and instructed to consume only the pre-packaged foods for the three days prior to each Test Visit. The details and isotopic abundances of the foods have been described in detail previously by Shoemaker et al. [31].

2.10. Statistical Analyses

All variables were assessed for distributional normality with Shapiro Wilks tests and analyzed for outliers with stem and leaf plots. Independent samples t-tests were used to compare

age, maturity offset, height, body mass, percent body fat, and fat-free mass. Three-way mixed factorial analyses of variance (ANOVAs) [drink (RDC vs. SDC) x sex (male vs. female) x time (15 vs. 30 vs. 45 vs. 60 vs. 75 vs. 90 vs. 105 vs. 120 vs. 135 vs. 150 vs. 165 vs. 180)] were used to compare mean CHO and fat oxidation rates, as well as cumulative CHO and fat oxidation. Three-way mixed factorial ANOVAs [drink (RDC vs. SDC) x sex (male vs. female) x time (15 vs. 30 vs. 45 vs. 60 vs. 75 vs. 90 vs. 105 vs. 120 vs. 135 vs. 150 vs. 165 vs. 180)] were used to compare mean endogenous and exogenous CHO oxidation rates, as well as cumulative endogenous and exogenous CHO oxidation. Three-way mixed factorial ANOVAs [drink (RDC vs. SDC) x sex (male vs. female) x time (60 vs. 120 vs. 180)] were used to compare the relative contributions of fat, endogenous, and exogenous CHO oxidation to total substrate oxidation at rest, during exercise, and post-exercise rest, expressed as cumulative area under the curve. Three-way mixed factorial ANOVAs [drink (RDC vs. SDC) x sex (male vs. female) x time (0 vs. 30 vs. 60 vs. 120 vs. 180)] were used to compare glucose and insulin responses. Significant interactions were decomposed with follow-up lower-order ANOVAs and Bonferroni-corrected dependent and/or independent samples t-tests. All statistical analyses were performed in IBM SPSS v. 25 (Chicago, IL, USA). An alpha of $p \leq 0.05$ was considered statistically significant for all comparisons.

3. RESULTS

Participant demographics are listed in Table 1. All participants were between the 10th and 95th percentiles for height, 5th and 95th percentiles for body mass, and 10th and 90th percentiles for body mass index. Boys were older, taller, and had greater body mass and fat-free mass ($p \leq 0.005$). Estimated peak height velocity was greater in the boys ($p < 0.001$), however, there were no differences among estimated maturity offset ($p = 0.406$). There was no evidence of a sex x

drink interaction for any comparisons ($p \geq 0.093$), therefore, data are presented collapsed across sex.

[TABLE 1 ABOUT HERE]

There was a significant drink x time interaction for total CHO oxidation rate ($p < 0.001$). Total CHO oxidation rate was greater after the RDC drink at 60 min ($p = 0.032$, Figure 3A). There were no other drink-related differences for total CHO or fat oxidation rate ($p \geq 0.146$, Figure 3A and 3B), or cumulative CHO and fat oxidation ($p \geq 0.202$, Figure 3C and 3D). After both drinks, total CHO oxidation rate was greater at 30, 45, and 60 min than 15 min ($p \leq 0.007$), and greater at 75, 90, 105, and 120 min than all other time points ($p < 0.001$, Figure 3A), and fat oxidation rate was greater at 75, 90, 105, and 120 min than all other time points ($p < 0.001$, Figure 3B). There were significant main effects for cumulative CHO and fat oxidation such that cumulative fat and CHO oxidation increased systematically across time collapsed across drink ($p < 0.001$, Figure 3C and 3D, respectively).

[FIGURE 3 ABOUT HERE]

There was a significant drink x time interaction for endogenous oxidation rate ($p < 0.001$, exogenous oxidation rate ($p < 0.001$), cumulative endogenous CHO oxidation ($p < 0.001$), and cumulative exogenous oxidation ($p < 0.001$). Endogenous CHO oxidation rate was greater after the SDC drink at 15 min ($p \leq 0.010$, Figure 4A). Cumulative endogenous CHO oxidation was greater after the SDC drink at 45 min ($p = 0.009$, Figure 4C). Exogenous CHO oxidation rate was greater for the RDC at 30, 45, 60, 75, 90, 105, 120, and 135 min ($p \leq 0.036$, Figure 4B), while cumulative exogenous CHO oxidation was greater for the RDC at 30, 45, 75, 90, 105, 120, 135, 150, 165, and 180 min ($p \leq 0.024$, Figure 4D). After both drinks endogenous CHO oxidation rate was greater at 30, 45, and 60 min than 15 min ($p \leq 0.023$), and greater at 75, 90,

105, and 120 min than all other time points ($p \leq 0.011$, Figure 4A). Exogenous CHO oxidation rate was greater at 75, 90, 105, and 120 min than all other time points after both drinks ($p \leq 0.025$, Figure 4B). After both drinks, there were systematic increases from 15 to 165 min for cumulative endogenous CHO oxidation ($p \leq 0.007$, Figure 4C). For the RDC drink, cumulative exogenous CHO oxidation remained the same from 15 to 75 min ($p \geq 0.102$) and increased systematically from 75 to 180 min ($p \leq 0.044$, Figure 4D). For the SDC drink, cumulative exogenous CHO oxidation remained the same from 15 to 75 min ($p \geq 0.158$), increased from 75 to 120 min ($p \leq 0.048$), and remained the same from 135 to 180 min ($p \geq 0.056$, Figure 4D).

[FIGURE 4 ABOUT HERE]

When exogenous CHO oxidation, endogenous CHO oxidation, and fat oxidation were expressed as a relative contribution to total substrate oxidation (expressed as cumulative area under the curve for 0-60 minutes (initial rest), 0-120 minutes (initial rest + exercise), and 0-180 minutes (initial rest + exercise + post-exercise rest) there was a significant main effect for exogenous CHO oxidation and significant drink x time interaction for endogenous CHO oxidation ($p < 0.001$). Exogenous CHO oxidation accounted for a greater proportion of total substrate oxidation during the entire study protocol for the RDC collapsed across time ($p \leq 0.019$, Figure 5). Endogenous CHO oxidation accounted for a greater proportion of total CHO oxidation for the SDC after the initial rest ($p = 0.028$, Figure 5A). Though not statistically significant, endogenous CHO oxidation increased from the RDC to the SDC after exercise and post-exercise rest ($p \geq 0.228$, Figure 5B and 5C), while fat oxidation increased from the RDC to the SDC after rest, exercise, and post-exercise rest ($p \geq 0.384$, Figure 5).

[FIGURE 5 ABOUT HERE]

There was a significant main effect for glucose, such that blood glucose was greater at 30 and 60 min than all other time points collapsed across drink ($p < 0.001$). There were no differences across time for insulin responses ($p = 0.181$), as well as no differences between drinks for glucose or insulin responses across the entire study protocol ($p \geq 0.088$, Figure 6). There was a trend ($p = 0.088$) for higher glucose at 60 and 120 minutes for the SDC. Evaluation of individual glucose responses (Figure 6C and 6D) demonstrated a bimodal distribution for glucose responses, with some participants having an initial rise at 30 minutes followed by their true peak at 60 minutes after consumption of the SDC (Figure 6D), compared to a unimodal distribution with a single glucose peak at 30 minutes after consumption of the RDC (Figure 6C).

[FIGURE 6 ABOUT HERE]

4. DISCUSSION

To our knowledge, the present study is the first to report endogenous and exogenous CHO oxidation, fat oxidation, glucose, and insulin responses from a fasted to a fed state at rest, during exercise, and post-exercise after consumption of slow versus rapid digesting CHOs in pre-pubertal males and females. The primary findings of this study demonstrated greater exogenous CHO oxidation (Figure 4B and 4D) and a greater proportion of total substrate oxidation from exogenous CHO oxidation (Figure 5) following consumption of the RDC drink compared to the SDC drink. Added sugar can largely contribute to the energy density of the diet and may promote weight gain as body fat in children [1,7,9,10,32]. However, recent studies have suggested that foods incorporating slowly digestible added sugars may have a significant health benefit on body weight and metabolic regulation [8,33]. Thus, the present study provides novel data suggesting that an SDC may promote greater endogenous substrate utilization in pre-pubertal children, which may have beneficial health impacts on energy intake and carbohydrate

regulation/metabolism during growth and development. Furthermore, the present results demonstrate the added benefit of exercise to promoting endogenous substrate utilization after consumption of an SDC blend.

The prevalence of metabolic dysfunction and obesity in children has increased significantly from 2000-2016, despite being a major public health initiative [34,35]. In fact, in 2015 the WHO released guidelines recommending that a decrease in rapid digesting sugar intake may help combat the worldwide childhood obesity epidemic [1]. In support of the WHO guidelines, previous studies in children have suggested that greater dietary sugar intake may lead to greater daily caloric consumption, lipogenesis, and subsequent weight gain as fat [7,9,10,32,36–38]. Thus, although previous studies in children and adults have generally examined the benefit of CHO beverages from a sports nutrition perspective [15–21,39,40], with the goal of increasing exogenous CHO oxidation to improve performance, it could be argued that this may in fact negatively affect long-term health in children. Interestingly, although these studies have primarily been interested in increasing exogenous CHO oxidation through consumption of RDCs, their findings have demonstrated that different blends of CHO may in fact change the proportion of energy oxidized from exogenous versus endogenous CHO sources. This may prove to be beneficial in combatting the metabolic issues related to obesity in children by influencing substrate utilization after meal ingestion. Rueda-Maza et al. [41] demonstrated that obese pre-pubertal children have no difficulty oxidizing the CHOs contained in a mixed meal. In fact, in the more obese children, a greater proportion of total CHO oxidation was accounted for with exogenous CHO oxidation compared with more lean children. Based on the results of the present study, it is possible that the SDC may be beneficial for obese children due to its effect on endogenous substrate utilization, though further research in obese children is

necessary. Therefore, increasing glycogen (i.e., endogenous CHO oxidation) turnover during periods of growth may be beneficial to maintain glucose homeostasis and energy balance, and reduce the risk of obesity later in life.

Previous studies in animals have suggested that consumption of SDCs may cause a shift toward greater reliance on endogenous substrate oxidation during the early stages of growth and development [11–14]. These studies showed greater endogenous fat oxidation after consumption of an SDC compared to an RDC. Most recently, Salto et al. [13] found that animals fed an SDC diet maintained a more balanced oxidation of endogenous CHO and fat for energy production, while the RDC diet primarily oxidized CHO. The study also suggested an RDC diet may promote preferential subcutaneous and visceral fat deposition compared to the SDC diet. Furthermore, SDC diets in animals may promote increases in muscle glycogen and maintenance of hepatic glycogen content [11–14]. We recommend caution when extrapolating these findings to the present study, since the present study was an acute study in healthy pre-pubertal children compared to a chronic early programming animal study, and the present study did not provide a source of exogenous fat (only CHO), meaning the present participants had to rely on CHO as the source of exogenous energy. Nonetheless, even though the present study design was different from previous studies in animals, there was a trend towards greater endogenous fat oxidation after consuming the SDC compared to the RDC (Figure 5). Thus, the present results are in agreement with the plausible hypothesis pointed out by previous animal studies [11–14].

Although not specifically comparing SDCs and RDCs, several studies in adults [15–20] found that consumption of CHO blends including glucose + fructose, glucose + sucrose, and glucose + fructose + sucrose resulted in greater exogenous CHO oxidation compared to consumption of a CHO drink containing glucose or sucrose alone during submaximal cycling.

Similarly, other studies in adults [39,40] reported greater exogenous CHO oxidation following consumption of a CHO blend of maltodextrin + fructose compared to a CHO drink of maltodextrin alone during submaximal cycling. In 10-14-year old males, Riddell et al. [21] reported similar exogenous and endogenous CHO oxidation after consumption of a CHO blend of glucose + fructose compared to a CHO drink containing glucose alone during 90 min of submaximal cycling. Collectively, these studies suggest that augmenting the blend of CHO consumed may change the proportion of CHO oxidized from exogenous vs. endogenous CHO sources during submaximal exercise.

As mentioned previously, studies in children and adults have primarily examined the benefits of CHO drinks from a sports nutrition perspective with the goal of improving exercise performance [15–21,39,40]. However, we are unaware of any previous studies in prepubertal children to include measurements of resting and exercise metabolism after consumption of an SDC or RDC blend to examine the health-related effects with regard to substrate utilization. For the pre-pubertal males and females in the present study, exogenous CHO oxidation rate ($\text{g}\cdot\text{min}^{-1}$) was greater during pre-exercise rest and exercise for the RDC drink compared to the SDC drink (Figure 4B), while cumulative exogenous oxidation (g) was greater for the RDC drink across the entire study protocol (Figure 4D). Furthermore, a greater proportion of total substrate oxidation was provided by exogenous CHO oxidation for the RDC drink during pre-exercise rest, exercise, and post-exercise rest (Figure 5), while a greater proportion of total substrate oxidation was provided by endogenous CHO oxidation for the SDC drink during pre-exercise rest (Figure 5A). Thus, the unique SDC and RDC blends in the present study present unique findings in pre-pubertal males and females, suggesting that an SDC may promote a greater proportion of substrate oxidation from endogenous sources. In fact, not only were these results true during the

resting period, as demonstrated previously in children [41], but also during the exercise protocol, providing new insight regarding the added value of exercise in conjunction with an SDC blend to promote endogenous substrate utilization and promote health-related outcomes. Furthermore, these results are in agreement with previous studies in animals [11,12,14], which preliminarily suggests that consumption of SDCs may be able to combat lipogenesis and promote healthy weight gain as children grow and develop. Therefore, the present findings suggest that a specifically designed SDC blend may promote endogenous substrate utilization, which may prove beneficial in combatting the childhood obesity epidemic, though longitudinal studies and studies involving obese children are necessary to better understand this. The present results suggest that the combination of an SDC and exercise may prove beneficial when the goal is to promote endogenous substrate utilization, particularly when considering the innate positive effects that exercise has on body composition and substrate utilization.

Previous studies in children and adults have reported conflicting information regarding glucose and insulin responses following consumption of different CHO blends. Possible explanations offered for conflicting outcomes among studies include differences in the physical activity intervention and/or the CHO amount consumed, and types studied. For example, many studies compared sucrose (table sugar) versus high fructose corn syrups [15–22,39,40]; however, both are compositionally identical (both are digested rapidly and absorbed as glucose and fructose). Several previous studies have reported similar glucose and insulin responses following consumption of glucose + fructose, glucose + sucrose, and glucose + fructose + sucrose compared to consumption of glucose or sucrose alone [16,18,20,39,40]. In the present study, the RDC blend provided 30 g of maltodextrin and sucrose, both of which are digested rapidly and absorbed, while the SDC blend provided both rapid and slow digested CHO sources, including

fermentable fiber. Unexpectedly, the results did not reveal a lower capillary glucose response after consumption of the SDC; in fact, there was a trend for higher glucose during the pre-exercise rest (Figure 6A). Since the low glycemic properties of isomaltulose, sucromalt, and fibersol individually or in combination are well characterized [4,5], a potential explanation could be that the postprandial glycemic peak may have occurred at an earlier time point (15 min) after consumption of the RDC as compared with SDC. In fact, the individual glucose responses show that after the SDC was consumed, a more bimodal distribution with the glycemic peak at 60 minutes appeared for several participants (Figure 6D). These results are more consistent with expectations that the SDC blend provided a slower release of glucose throughout the study protocol. However, further research is necessary to understand the unique influence an SDC drink may have on glucose and insulin responses at rest, during exercise, and during post-exercise rest in pre-pubertal children.

Although this study provides unique data regarding endogenous and exogenous substrate oxidation in prepubertal males and females there are limitations. First, the homogenous nature of our sample (born full-term, normal body mass, normal height, Tanner stage 1, etc.) limits our interpretation to be applied to healthy, prepubertal children. Future studies including a broader more diverse sample, including comparisons of healthy vs. obese children, may present clarifying information regarding the ability for an SDC blend to combat childhood obesity. Additionally, future studies quantifying total dietary intake throughout the study protocol may provide further insight regarding the role of an SDC blend as a part of a routine diet. Furthermore, although the present study compared resting to exercise metabolism, future studies comparing endogenous and exogenous substrate oxidation between a group receiving an exercise intervention versus a non-exercise group may provide further information regarding the

influence of exercise on substrate oxidation after consumption of an SDC or RDC blend.

Nevertheless, the present study presents novel information that suggests an SDC blend may positively influence substrate oxidation during rest and exercise in prepubertal children.

5. CONCLUSION

In conclusion, the results of the present study present unique findings suggesting that consumption of an SDC drink may alter CHO oxidation in pre-pubescent children. Specifically, the SDC drink consumed in the present study may promote a greater proportion of CHO oxidation from stored glycogen (endogenous sources) compared to consumption of an RDC drink. In conjunction with previous studies [34,35], and recommendations from the WHO [1], promoting endogenous CHO oxidation at rest, during exercise, and during post-exercise rest may help combat the growing prevalence of childhood obesity. Thus, the unique SDC blend in the present study may present an alternative CHO blend that could be consumed to promote the breakdown and oxidation of stored glycogen, which is critical to maintain energy balance as children grow and develop [41], and may be an effective action for targeting key risk factors in future childhood obesity prevention programs. Future research may consider examining the influence of an SDC blend on substrate utilization in obese children. Furthermore, since the present study only involved an acute measurement of metabolism after ingestion of an SDC and RDC drink, further research involving longitudinal measurements of metabolism after ingestion of SDCs and RDCs in normal weight and obese children are needed to better understand the role of SDCs in combatting the childhood obesity epidemic.

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CONFLICT OF INTEREST

Current disclosures for JTC: From 2001-present, Dr. Cramer's research has been externally funded by 22 separate project grants from Abbott Nutrition, Nebraska Beef Council, Nebraska Extension, Stepan Lipid Nutrition, Rock Creek Pharmaceuticals, General Nutrition Corporation, Experimental & Applied Sciences, Nutricia, and the University of Nebraska Agriculture Research Division with funds provided by the Hatch Act (Agency: U.S. Department of Agriculture, National Institute for Food and Agriculture: Accession No: 1000080; Project No: NEB-36-078). From 2010-2013, Joel was an expert witness or expert consultant in 10 separate legal matters representing defendants Vital Pharmaceuticals, MusclePharm, and Celsius. From 2008-2016, Dr. Joel T. Cramer was a paid consultant for either Abbott Nutrition, General Nutrition Center, ErgoGenix/ErgoPharm, and/or Corr-Jensen Labs. From 2018-present, Joel has served as a paid consultant for Regeneron Pharmaceuticals. Jose M. López-Pedrosa and Ricardo Rueda are employees of Abbott Nutrition and stockholders.

CRedit AUTHORSHIP CONTRIBUTION STATEMENT

Zachary M. Gillen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft. **Vikkie A. Mustad:** Conceptualization, Formal analysis, Methodology, Resources, Writing – review & editing. **Marni E. Shoemaker:** Formal analysis, Visualization, Writing – review & editing. **Brianna D. McKay:** Data curation, Investigation, Project administration, Writing – review & editing. **Todd J. Leutzinger:** Data curation, Investigation, Project administration, Writing – review & editing. **Jose M. Lopez-Pedrosa:** Conceptualization, Methodology, Resources, Writing – review & editing. **Ricardo Rueda:** Conceptualization, Methodology, Resources, Writing – review & editing. **Joel T. Cramer:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft.

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FIGURE CAPTIONS

Figure 1. Flow diagram for assessment, randomization, allocation, and analysis of study participants. RDC = rapid digesting carbohydrate, SDC = slow digesting carbohydrate

Figure 2. Assessments performed at the Pre-screening Visit, Baseline Visit, and a visual depiction of Test Visits 1 and 2. Br = breath test, Bl = blood sample.

Figure 3. Means $\pm$ standard errors for total carbohydrate (CHO) and fat oxidation rate (A and B, respectively), and cumulative CHO and fat oxidation (C and D, respectively) for the rapid digesting carbohydrate (RDC) and slow digesting carbohydrate (SDC) drinks. * Indicates difference between drinks, † indicates greater than 15 min, ‡ indicates greater than 135, 150, 165, and 180 min, ¥ indicates greater than all other time points, § indicates systematic increases.

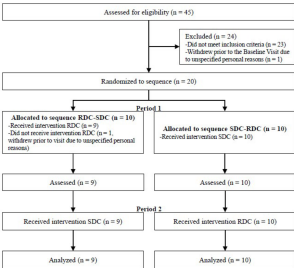
Figure 4. Means $\pm$ standard errors for endogenous and exogenous carbohydrate (CHO) oxidation rate (A and B, respectively), and cumulative endogenous and exogenous CHO oxidation (C and D, respectively) for the rapid digesting carbohydrate (RDC) and slow digesting carbohydrate (SDC) drinks. * Indicates difference between drinks, † indicates greater than 15 min, ‡ indicates greater than 135, 150, 165, and 180 min, ¥ indicates greater than all other time points, § indicates systematic increases for RDC, § indicates systematic increases for SDC.

Figure 5. Means $\pm$ standard errors for the relative contributions to total substrate oxidation for fat, endogenous carbohydrate oxidation (Glycogen), and exogenous carbohydrate oxidation

(Diet) at the end of A) rest, B) rest + exercise, and C) rest + exercise + post-exercise rest. *

Indicates a significant difference between drinks.

Figure 6. Means $\pm$ standard errors for mean glucose and insulin responses (A and B, respectively) after consuming the rapid digesting carbohydrate (RDC) and slow digesting carbohydrate (SDC) drinks, as well as individual glucose responses after consuming the RDC and SDC (C and D, respectively). * Indicates greater than 0, 120, and 180 min.



Pre-screening Visit

Assessment of Participant Eligibility
Completion of Youth Assent
Completion of Parental Consent
Completion of PAR-Q+ 2015



7-14 days

Baseline Visit

Anthropometrics
Body Composition
Submaximal Cycle Ergometer Test



7-14 days

Delivery of pre-participation health questionnaire three days prior to Test Visit 1

Test Visit 1



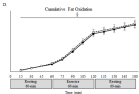
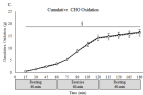
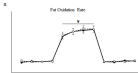
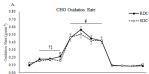
7-14 days

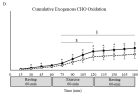
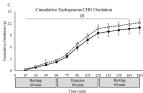
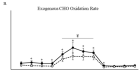
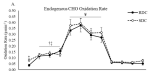
Delivery of pre-participation health questionnaire two days prior to Test Visit 2

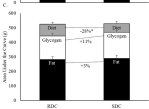
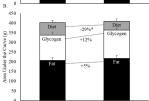
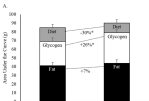
Test Visit 2



■ 15-min averages of Indirect Calorimetry







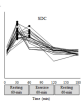
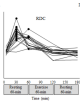
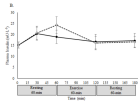
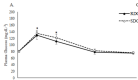


Table 1. Mean $\pm$ standard errors for age, maturity offset, height, body mass, percent body fat, and fat-free mass.

	Age (years)			Maturity Offset (years)			Height (cm)			Body Mass (kg)			Percent Body Fat			Fat-free Mass		
Boys (n = 10)	11.00	$\pm$	0.39*	-3.05	$\pm$	0.26	148.96	$\pm$	3.07*	41.10	$\pm$	2.90*	22.80	$\pm$	2.78	26.85	$\pm$	1.20*
Girls (n = 9)	8.56	$\pm$	0.28	-2.75	$\pm$	0.28	135.40	$\pm$	0.89	29.82	$\pm$	0.75	23.43	$\pm$	7.37	19.16	$\pm$	0.53

* Indicates a significant difference between boys and girls ($p \leq 0.05$)